A witness ending at the crossing

Human Compression project: L044 continuation

We first record three local facts with string commutations. Let $Q = \langle \rho_m, \dots, \rho_y \rangle$ be transitive on W, with every displayed mark active. Suppose the full local j-omission has exactly two orbits A, B, joined by a unique j-edge $\{a, b\}$. Put

$$L = Q_{<j}, \quad R = Q_{>j}, \quad M = Q_{>m}, \quad h = j - m, \qquad C = a^L \sqcup b^L.$$

The actual omission sides are transitive under the commuting product LR. Marks at distance at least two from j fix a, b. The acquired anchored counts give $|C| \geq h + 2$. For $h \geq 1$, let $L' = \langle \rho_{m+1}, \dots, \rho_{j-1} \rangle$. The strict proper-prefix bound, applied in reverse order to the left action, gives

$$|a^{L'}| + |b^{L'}| \geq h + 1. \tag{1}$$

Indeed a side where the left active length is h contributes at least h points to this sum; the other contributes one. For $h = 1$, $L' = 1$ and the same assertion is immediate. Both sets in (1) lie in a single M-orbit, since the crossing belongs to M and joins their anchors. No intersection property in a restricted action is assumed.

Lemma 1 (The minimum left union)**.** *Suppose $y > j$ and $|C| = h + 2$.*

1. *No nontrivial M-orbit on which R acts trivially meets C.*
2. *If M fixes a point $z \in C$, its orbits on W are exactly $\{z\}$ and a transitive complement. The complement has nontrivial R-action.*

Proof. If $h = 0$, then $M = R$ and $C = \{a, b\}$. The first assertion is immediate. A fixed anchor has a singleton actual omission side, while R is transitive on the other. The latter action is nontrivial because $y > j$. This proves the second assertion.

Let $h \geq 1$. One left orbit has size $h+1$ and the other is a singleton, since one left active length is h. Name them $A_0 = a^L$ and $\{b\}$. The set $P = a^{L'}$ has size at least h. Thus $E = P \cup \{b\}$ lies in a single M-orbit and $|C \setminus E| \leq 1$.

If an R-trivial M-orbit met E, it would contain a, b. Every element of R would fix both anchors and centralize L, hence fix C pointwise. The actual LR-orbits would then be precisely A_0 and $\{b\}$, making $W = C$ and R trivial on W, a contradiction. Hence such an orbit meeting C can meet it only at the possible remaining point z. Since R fixes z and centralizes L, it fixes $A_0 = z^L$, so the actual side A is A_0. The orbit cannot leave A: its only possible exit is the unique crossing at a, already in E. Inside A it avoids P and therefore has at most one point. This proves assertion 1.

For assertion 2, the fixed point z is neither a nor b, since $\rho_j \in M$. It belongs to $A_0 \setminus P$. Again R fixes A_0, making it the actual side A. The left factor fixes b and centralizes R, so the other actual side is $B = b^R$. Equality forces $P = A \setminus \{z\}$. The surviving left marks connect P, the right marks connect B, and the crossing joins them. Thus their complement to z is one M-orbit. All nontrivial R-action is on this complement. □

Lemma 2 (One extra point in the left union)**.** *Suppose $h \geq 1$, $y > j$, $|C| = h + 3$, and M fixes some $z \in C$. Then every nontrivial M-orbit has nontrivial R-action and meets C. Every M-fixed point belongs to C.*

Proof. First let $h \geq 2$. Name a left orbit of active length h as $A_0 = a^L$. Either

$$(|A_0|, |b^L|) = (h+1, 2) \quad \text{or} \quad (h+2, 1).$$

In the first case only the nearest left mark ρ_{j-1} can act on the two-point orbit b^L: all farther marks fix b and hence fix that pair. The nearest mark exchanges the pair and survives in M. Thus the fixed point z is in A_0. In the second case the singleton b is moved by $\rho_j \in M$, so again $z \in A_0$.

The factor R fixes z and centralizes L, hence fixes A_0 pointwise. Consequently the actual side A is A_0. The surviving left orbit $P = a^{L'}$ has at least h points and does not contain z.

When $|A_0| = h+1$, we have $P = A \setminus \{z\}$. On the other actual side, all left marks except ρ_{j-1} fix b and centralize R, so fix all of $B = b^{LR}$. The surviving nearest left mark together with R is therefore transitive on B. The crossing joins P to B, leaving precisely the fixed point z. This complementary orbit has nontrivial R-action because R fixes A and is nontrivial on W.

When $|A_0| = h+2$, the left factor fixes b, so $B = b^R$ is transitive under R. The M-orbit containing P and B has nontrivial R-action and meets C. At most two points of A lie outside it, one being the fixed point z. There is consequently no room for another nontrivial M-orbit. Any other point outside this orbit is fixed and belongs to $A \subset C$. This proves the result for $h \geq 2$.

For $h = 1$, the single left involution has C equal to two two-point orbits, say $\{a, a_1\}$ and $\{b, b_1\}$. The fixed point of $M = \langle \rho_j, R \rangle$ is not a, b; name it a_1. Commutation makes R fix both a, a_1, so this pair is the complete actual side A. On the other side B, the transitive commuting product LR has either one or two R-orbits. Each meets $\{b, b_1\}$; if there are two, L exchanges them. None is a singleton: if R fixed b, it would fix b_1 too, and the full actual sides would be the two pairs, with R trivial on W. The crossing attaches a to the R-orbit of b. The action of ρ_j may join that orbit to the other R-orbit in B, if present; it cannot create any new orbit. Thus every nontrivial M-orbit contains a nontrivial R-orbit, meets C, and the only fixed point is a_1. This proves the lemma. □

Lemma 3 (The shorter witness at its minimum)**.** *Suppose instead $y = j$, $h \geq 1$, $|W| = h+2$, and M is intransitive. Then M has one singleton and one transitive orbit of size $h+1$.*

Proof. Here $R = 1$ and the actual omission sides are left orbits of sizes $h+1$ and one. The set in (1), joined by the crossing, lies in one M-orbit of size at least $h+1$. Intransitivity leaves exactly one point outside it, which is fixed. This proves the assertion without classifying the individual marks. □

Now retain all complete actual odd-degree head and globally paired-tail hypotheses, and choose $j \geq 5$ by the strong simultaneous-witness theorem. Write $H = G_{\geq 5}$, $K = G_{>5}$, $\sigma = \rho_4$, and $J = G_{>j}$. Denote the head attachment point by β, to distinguish it from local crossing endpoints. Thus

$$D = \{1, \ldots, 6\}, \quad \rho_3 = (6\ \beta)\delta, \quad H|_{D \cup \{\beta\}} = 1, \quad \sigma|_D = 1.$$

The actual anchor $z_0 = \beta^\sigma$ is outside the seven head points and fixed by K. Every K-intransitive witness orbit is fixed pointwise by the odd involution δ.

Theorem 4. *Suppose two distinct K-intransitive witness orbits O_1, O_2 have last active labels $j < y$, respectively. There is a nonterminal index s with*

$$|\operatorname{Fix}(G_{>s})| \geq 2s + 1.$$

For $j \geq 6$ one may take $s = j+1$. For $j = 5$ one may take $s = 6$ or $s = y$.

Proof. Put $s = j+1$ for the moment. If $s = r-1$, then $y = r-1$ also. The seven head points and the two complete witness orbits are fixed by δ and have total size at least

$$7 + (j-3) + (j-2) = n - 3.$$

An odd involution supported on at most three points is a literal transposition. The complete acquired exterior-pair obstruction excludes it under the present actual head and global pair hypotheses. Therefore $s < r-1$.

Let $h = j - 5$, and in O_2 set $L = G_{\{5,\dots,j-1\}}$ and $C = a^L \sqcup b^L$ as above. Both C and O_1 are fixed by every mark after s: those marks fix a, b and commute with L. The acquired bounds give

$$|O_1| \geq h + 2, \quad |C| \geq h + 2, \quad |F| \geq 2j + 1, \qquad F = \mathrm{Fix}(G_{>s}). \tag{2}$$

If $|F| \geq 2j+3$, this is the required separator. Otherwise F has at most one point beyond the lower bound in (2). Its disjoint constituents are the seven head points, O_1, C, and possibly one further point. The involution σ preserves F and permutes actual K-orbits, since it centralizes both K and $G_{>s}$. The actual anchor z_0 is in F.

We repeatedly use the following explicit invariance check. A union consisting of the seven head points and some whole witness orbits is preserved by the first three marks, by ρ_3, and by H: the first three preserve D; ρ_3 exchanges $6, \beta$ and fixes the witnesses through δ; and H preserves each of its orbits and fixes the head. Once this union is also σ-invariant, it is G-invariant. It is proper whenever δ fixes it pointwise, since δ is odd and thus nonidentity. We will also use that a witness alone is G-invariant if σ preserves it: all earlier marks fix it, and H preserves it.

Both lower bounds in (2) are equalities, with zero or one further point q. First assume $h \geq 1$. The shorter witness lemma makes the K-orbits of O_1 a singleton $\{z_1\}$ and a transitive orbit T_1 of size $h+1 \geq 2$. The subgroup J fixes T_1. By the minimum-left-union lemma, its nontrivial K-orbit image T_1^σ cannot meet C. It cannot meet the fixed head points or fit in the one possible extra point. Hence $T_1^\sigma = T_1$.

The anchor z_0 cannot be z_1: that would make the head together with O_1 invariant, contrary to the preceding check. Thus z_0 is in C or is the extra point q.

If $z_0 \in C$, the minimum-left-union lemma makes the K-orbits of O_2 a singleton $\{z_0\}$ and a transitive complement T_2 with nontrivial J-action. If σ moved T_2 to a distinct orbit, the images of the $h + 1 \geq 2$ points of $C \setminus \{z_0\}$ would all have to lie outside O_2 in F. They cannot lie in T_1, on which J is trivial, or among K-fixed points. Only the one possible extra point remains, contradicting injectivity. Thus T_2 is preserved.

If the extra point q is K-fixed, it is also H-fixed and δ-fixed. Indeed both ρ_5 and δ preserve F; the former preserves its head, O_1 and C, and the latter fixes those sets pointwise. They must therefore fix the sole remaining point q. Adjoin q to the union of the head and both witnesses when it is K-fixed. Its nontrivial K-orbits are preserved by σ, and its fixed points are exactly the K-fixed points of F, also permuted by σ. This union is consequently G-invariant and pointwise δ-fixed, a contradiction.

If instead $z_0 = q$, then q is H-fixed and δ-fixed by the same argument. Since σ preserves T_1, it cannot fix z_1, or O_1 alone would be invariant. The point z_1^σ must be a K-fixed point $w \in C$: the head attachment β and q already form a σ-pair, and D is fixed. Apply the minimum-left-union lemma to w. The resulting complementary K-orbit of O_2 is again preserved by σ, since its at least two points in C cannot map to J-trivial T_1 or to fixed points. The head, both witnesses and q now form the same forbidden invariant union. This finishes this distribution for $h \geq 1$.

The shorter witness has one extra point. Here $|O_1| = h + 3$, $|C| = h + 2$ and there is no additional point in F. This is impossible for $h = 0$, since the sole active involution ρ_5 cannot be transitive on three points. For $h \geq 1$, every nontrivial K-orbit of O_1 is J-trivial. Its σ-image cannot meet C by the first local lemma, and cannot meet the fixed head. Thus σ permutes all the nontrivial K-orbits of O_1 within O_1.

If the actual anchor is in C, use it as a fixed point w there. Otherwise it lies in O_1. If every fixed point of O_1 were also sent into the head or O_1, the head with O_1 would be invariant. Consequently some fixed point of O_1 must be sent to a K-fixed point $w \in C$. In either case the minimum-left-union lemma gives one singleton w and a J-nontrivial transitive complement on O_2. Its σ-image must stay in O_2: it meets F through $C \setminus \{w\}$, while every nontrivial K-orbit in the rest of F is J-trivial. All nontrivial K-orbits of the union of the head and both witnesses are now permuted inside that union. Its fixed points are exactly the fixed points of K in F. The union is therefore invariant and pointwise δ-fixed, the same contradiction.

The left union has one extra point. Here $|O_1| = h + 2$, $|C| = h + 3$, with no other point in F. Necessarily $h \geq 1$, since for $h = 0$ the set C consists of exactly the two crossing endpoints. The shorter witness has K-orbits $\{z_1\}, T_1$, with $|T_1| = h + 1$ and J trivial on T_1. Put $E = a^{L'} \cup b^{L'}$ inside C. By (1), $|E| \geq h + 1$. Its containing K-orbit has nontrivial J-action: otherwise J would fix a, b and centralize L, forcing it to be trivial on all of O_2.

If $h \geq 2$ and σ moved T_1, its image would have to lie in $C \setminus E$, which has at most two points. This contradicts $|T_1| = h + 1 \geq 3$. If $h = 1$, the four points of C form two L-pairs $\{a, a_1\}$, $\{b, b_1\}$. The same image would have to be precisely $\{a_1, b_1\}$. Its J-triviality and commutation with L would make J fix a, b as well, again making J trivial on O_2. Hence σ preserves T_1 in every case.

The actual anchor cannot be z_1, by invariance of the head and O_1, so it is in C. Apply the one-extra-point local lemma. Every nontrivial K-orbit of O_2 meets C and has nontrivial J-action. Its σ-image therefore cannot leave O_2, since the rest of F has only fixed points and J-trivial T_1. All fixed points of K on O_2 are in C. Thus σ preserves the union of the head and both witnesses, including its fixed points. This is the same forbidden invariant union. All cases with $j \geq 6$ are excluded.

It remains to handle $j = 5$. Only the first distribution is possible: O_1 has two points, both fixed by K, $C = \{a, b\}$, and there is at most one additional point q in F. The involution δ fixes F pointwise. If q is K-fixed, it is also H-fixed: ρ_5 preserves F, the head, O_1, and the pair C, so fixes its possible remaining point.

We show that some point $w \in C$ is fixed by K, unless an invariant set already gives a contradiction. If the actual anchor lies in C, take it. If it lies in O_1, its other point must be sent by σ outside the head and O_1, or that union would be invariant. Its image is K-fixed. If it were q, the head, O_1 and q would be invariant, since q is H-fixed and δ-fixed. Hence the image lies in C. Finally, if the actual anchor is q, then β, q form a σ-pair. The set O_1 cannot be preserved by σ, or it alone would be invariant. A point of O_1 must therefore be sent to a K-fixed point in C. This proves the assertion in every case.

The minimum-left-union lemma now gives K-orbits $\{w\}$ and T on O_2. The complete local degree bound gives $|T| = |O_2| - 1 \geq y - 4$. If σ preserved T, the head, both witnesses and q when K-fixed would form the same forbidden invariant union. Thus $V = T^\sigma$ is a distinct actual K-orbit. The image of the other point of C belongs to $V \cap F$. It cannot lie in the fixed head or O_1, and cannot be in O_2, whose sole nontrivial K-orbit is T. Consequently the extra point q exists and belongs to V. Since δ fixes q and centralizes K, it fixes all of V pointwise.

The sets $D \cup \{\beta\}$, O_1, O_2 and V are disjoint. All are fixed by δ and by every mark after y; for V the latter assertion follows from commutation with σ. Their total size is

$$7 + 2 + (|T| + 1) + |T| \geq 2y + 2. \tag{3}$$

If $y = r - 1$, this leaves at most one point for the nontrivial odd involution δ, a contradiction. Thus $y < r - 1$, and (3) gives the required separator at $s = y$. The theorem follows. □

Attribution and boundary. The simultaneous witness choice, actual head anchor, strict anchored-prefix count, local degree bound and exterior-pair obstruction are acquired. The present proof supplies the local minimum and one-extra-point arguments and accounts for the action of ρ_4 on the entire fixed set. The two witnesses are explicitly required to be K-intransitive. A second witness with transitive K-action and the full global attachments remain outside this theorem. No whole-original, question, numerical or formal credit is claimed.